Robust estimation in partially nonlinear models.

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This is a joint work with

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Outline

1. Nonlinear models
2. Partially Nonlinear models
3. Theoretical results
4. Simulation Studies
5. Real example

Nonlinear models

Nonlinear models are used either by empirical or by theoretical considerations that suggest a functional form or as an attempt to express a known nonlinear relationship between the response and the predictor variables.

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Sometimes, the nonlinear trend can be transformed in order to obtain a linear model, however, there are several situations in which the transformation does not keep the main features of the original problem.

Nonlinear models

The nonlinear model assume that

$$y = g(\mathbf{x}, \boldsymbol{\beta}_0) + \sigma\varepsilon$$

where

- y is the dependent variable
- $\mathbf{x} \in \mathcal{X} \subset \mathbb{R}^k$
- $\boldsymbol{\beta}_0 \in \mathbb{R}^p$
- $\sigma \in \mathbb{R}_{\geq 0}$
- error term is independent of the covariates $\mathbf{x}$.
- We assume that the regression function g is known

In the classical setting, it is also assumed that $E(\varepsilon) = 0$ and $V(\varepsilon) = 1$.

Nonlinear models

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Classical inference procedures are based on the least squares method.

It is well known that the least squares estimator may turn to be very unstable in the presence of anomalous data and so, robust methods are welcome.

Nonlinear models

Some of the robust proposals introduced for linear regression have been extended to nonlinear models.

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- Fraiman (1983) proposed a general $M-$estimate
- Sromberg and Ruppert (1992) addressed the breakdown point concept in the nonlinear regression.
- Sakata and White (2001) extended the use of $S-$estimators
- Fasano (2009) derived the asymptotic theory of $MM-$ and $\tau-$ estimators.
- Bianco and Spano (2017) studied a family of weighted estimators that allows possible missing responses.
- etc

Nonlinear models: Examples

The air quality data set

Cleveland (1985) considers air quality measurements in a specific area of New York in order to predict the amount of Ozone from the wind speed.

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The 153 observations consist in daily readings of four air quality variables between 1 May 1973 and 30 September 1973.

In this data set, 37 observations of the response (Ozone) are missing.

Cleveland looked for the relationship between Ozone (ppb) and wind speed (mph) detecting a non linear trend.

Nonlinear models: air quality data set

Bianco and Spano (2017), fitted an exponential growth model with regression function given by

$$g(V, \beta_1, \beta_2, \beta_3) = \beta_1 \exp(\beta_2 V) + \beta_3 \tag{1}$$

to explain the relationship between ozone and wind speed.

They took into account the missing data.

Also, they analized the data detecting five outliers in the sample

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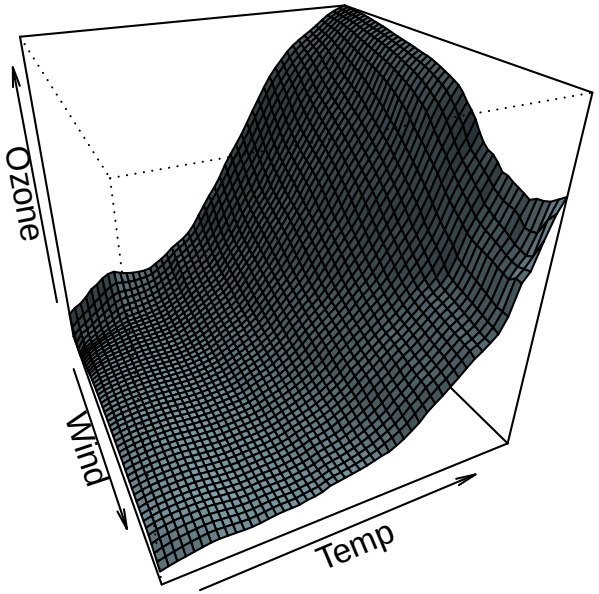
We analyse if the temperature could contribute to explain ozone levels.

We compute a robust nonparametric estimator explaining the ozone as a nonparametric function of the variables temperature and wind speed.

Nonlinear models: Examples

We can observe that ozone depends clearly on different conditions of temperature.



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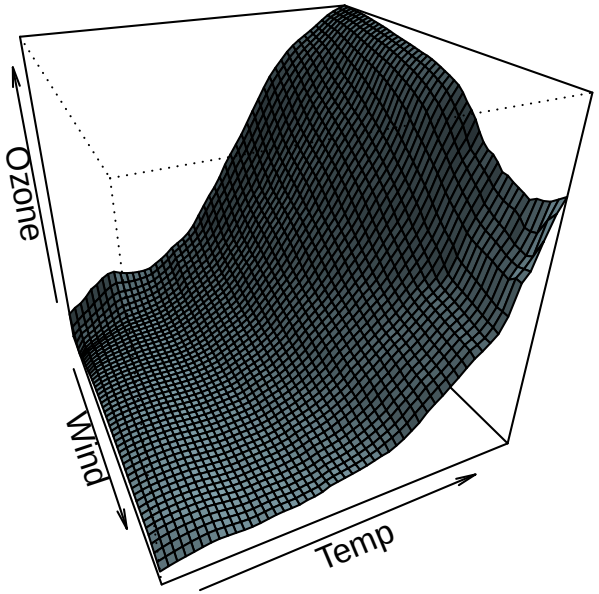

Therefore, it seems natural develop a model that include the temperature in order to explain ozone.

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Semi-parametric regression models are very popular in practical applications

Partially nonlinear models

Let y be a response variable and $(\mathbf{x}, t)$ be its covariates, we consider the following partially nonlinear model:

$$y = g(\mathbf{x}; \boldsymbol{\beta}_0) + \eta_0(t) + \varepsilon$$

where

- $g(\mathbf{x}; \boldsymbol{\beta})$ is a pre-specified function
- $\boldsymbol{\beta}_0$ is a $p-$dimensional unknown parameter vector
- η_0 is an unknown smooth function
- ε is the random error independent of the covariate vector $\mathbf{x}$

Partially nonlinear models

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- Some estimation procedures:
  - Li and Nie (2008) introduced the profile nonlinear least squares approach and the linear approximation method.
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All these proposal are closely related to the classical least squares technique.

Partially nonlinear models

- Robust estimators for partially linear models was proposed in Bianco y Boente (2004).
- Robust estimators for generalize partially linear models have been considered for instance by Boente et al. (2006) and by Boente and Rodriguez (2010).

Partially nonlinear models

Step 1: For each τ and $\boldsymbol{\beta}$, we define $\widehat{\eta}_{\boldsymbol{\beta}}(\tau)$ as solution of

$$arg \min_{a \in \mathbb{R}} \sum_{i=1}^{n} \rho \left(\frac{y_i - g(\mathbf{x}_i, \boldsymbol{\beta}) - a}{\widehat{s}_{\beta}(\tau)} \right) w_i(\tau),$$

where $w_i(\tau) = K\left(\frac{\tau - t_i}{h}\right)$ are the kernel weights on t_i, with K a kernel function and h a smoothing parameter and $\widehat{s}_{\beta}(\tau)$ is a local robust estimator for the scale.

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where $w_i(\tau) = K\left(\frac{\tau - t_i}{h}\right)$ are the kernel weights on t_i, with K a kernel function and h a smoothing parameter and $\widehat{s}_{\beta}(\tau)$ is a local robust estimator for the scale. For instance, $\widehat{s}_{\beta}(\tau)$ can be taken as the MAD with respect to the conditional distribution

$$\widehat{F}_{\beta}(v|t = \tau) = \sum_{i=1}^{n} w_i(\tau) I_{(-\infty, v]}(v_{i,\beta}) \left(\sum_{j=1}^{n} w_j(\tau) \right)^{-1},$$

where $v_{i,\beta} = y_i - g(\mathbf{x}_i, \boldsymbol{\beta})$.

Partially nonlinear models

Step 2: For each β compute the residuals $\widehat{r}_i(\beta) = y_i - g(\mathbf{x}_i, \beta) - \widehat{\eta}_\beta(t_i)$, and the estimator of β_0 as

$$\widehat{\beta} = arg \min_{\beta} \sum_{i=1}^{n} \rho_1 \left(\frac{\widehat{r}_i(\beta)}{s_n} \right) W(\mathbf{x}_i).$$

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$W(x)$ controls high leverage points in the $\mathbf{x}$ and where $s_n = \min_{\beta} \widehat{s}_0(\beta)$ is a M-scale estimate and $\hat{s}_0(\beta)$ is the solution in s_0 of

$$\frac{1}{n} \sum_{i=1}^{n} \rho_0 \left(\frac{\widehat{r}_i(\beta)}{s_0} \right) = b,$$

with $\rho_1 \leq \rho_0$.

Partially nonlinear models

Step 3: Define the estimate of the regression function η_0 as $\widehat{\eta}(\tau) = \widehat{\eta}_{\widehat{\boldsymbol{\beta}}}(\tau)$.

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In all the estimations steps , possible choices for the score function could be the Huber or the bisquare functions

Theoretical results: consistency

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- $\widehat{\beta} \xrightarrow{a.s.} \beta_0$
- $\|\widehat{\eta}_{\widehat{\beta}} - \eta_0\|_\infty \xrightarrow{a.s.} 0.$

Theoretical results: asymptotic distribution

Under some assumptions we have that

- $\sqrt{n}\left(\widehat{\beta} - \beta_0\right) \xrightarrow{\mathcal{D}} \mathbf{N}\left(0, A^{-1}\Sigma A^{-1^T}\right)$

Theoretical results: asymptotic distribution

Under some assumptions we have that

- $\sqrt{n}\left(\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0\right) \xrightarrow{\mathcal{D}} \mathbf{N}\left(0, A^{-1}\Sigma A^{-1^T}\right)$
  - $A = \frac{1}{\sigma_0}\mathbb{E}\left[\Psi_1^{'}\left(\frac{\epsilon}{\sigma_0}\right)\right]\mathbb{E}\left[W(\mathbf{x})z(\mathbf{x}, t, \boldsymbol{\beta}_0)z(\mathbf{x}, t, \boldsymbol{\beta}_0)^T\right]$
  - $\Sigma = \mathbb{E}\left[\Psi_1^2\left(\frac{\epsilon}{\sigma_0}\right)\right]\mathbb{E}\left[W^2(\mathbf{x})z(\mathbf{x}, t, \boldsymbol{\beta}_0)z(\mathbf{x}, t, \boldsymbol{\beta}_0)^T\right].$

$$
\begin{aligned}
z(\mathbf{x}, t, \boldsymbol{\beta}) &= \nabla_{\boldsymbol{\beta}}(g(\mathbf{x}, \boldsymbol{\beta}) + \eta_{\boldsymbol{\beta}}(t))\Big|_{\boldsymbol{\beta}} \\
\Psi_1 &= \rho_1^{'}
\end{aligned}
$$

Simulation Studies

We simulate the next model

$$y_i = (\mathbf{x}_i^T \boldsymbol{\beta}_0)^2 + \sin(2\pi t_i) + \varepsilon_i,$$

with $\boldsymbol{\beta}_0 = (1, 2, 1.5)^T$ and $\eta_0(t) = \sin(2\pi t)$.
The variables are distributed as follows

- $\mathbf{x}_i \sim N_3(\mathbf{0}, \mathbf{I_3})$
- $t_i \sim U[0, 1]$
- $\epsilon \sim N(0, 1)$.

- We generate $n = 100$ samples.
- The process is repeated $NRep = 1000$.

Simulation Studies

Contamination schemes

C_0: $\epsilon \sim N(0,1)$.

C_1: $\epsilon \sim t_1$, is a $\mathcal{T}$de Student with one degree of freedon.

C_2: $\epsilon \sim 0.9N(0,1) + 0.1N(0,10^2)$.

C_3: $\epsilon \sim t_1$ and replaced 10% of the values of $\mathbf{x}$ with observations equals $(5,5,5)$.

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Simulation Studies

- In the smoothing procedure,
  - we consider the Epanechnikov kernel $K(t) = \frac{3}{4}(1-t^2)I_{[-1,1]}(t)$
  - we take into account several bandwidths, but we will only show $h = 0.3$
- In **Step 1**, we consider
  - the Huber's function ρ_k with $k = 1,345$
  - the local MAD as a local scale $\widehat{s}_\beta(t)$
- In **Step 2**, we choose
  - the functions ρ_0 and ρ_1 belong to Tukey's bisquare family.
  - the weigths

$$W(\mathbf{x}) = \left(1 - \left(\frac{(\mathbf{x}-\hat{\mu})^T\hat{\Sigma}^{-1}(\mathbf{x}-\hat{\mu})}{\chi^2_{3,0.975}}\right)^2\right)^2 I_{(|(\mathbf{x}-\hat{\mu})^T\hat{\Sigma}^{-1}(\mathbf{x}-\hat{\mu})|\leq\chi^2_{3,0.975})}$$

with $\hat{\mu}$ and $\hat{\Sigma}$ robust estimators.

Simulation Studies

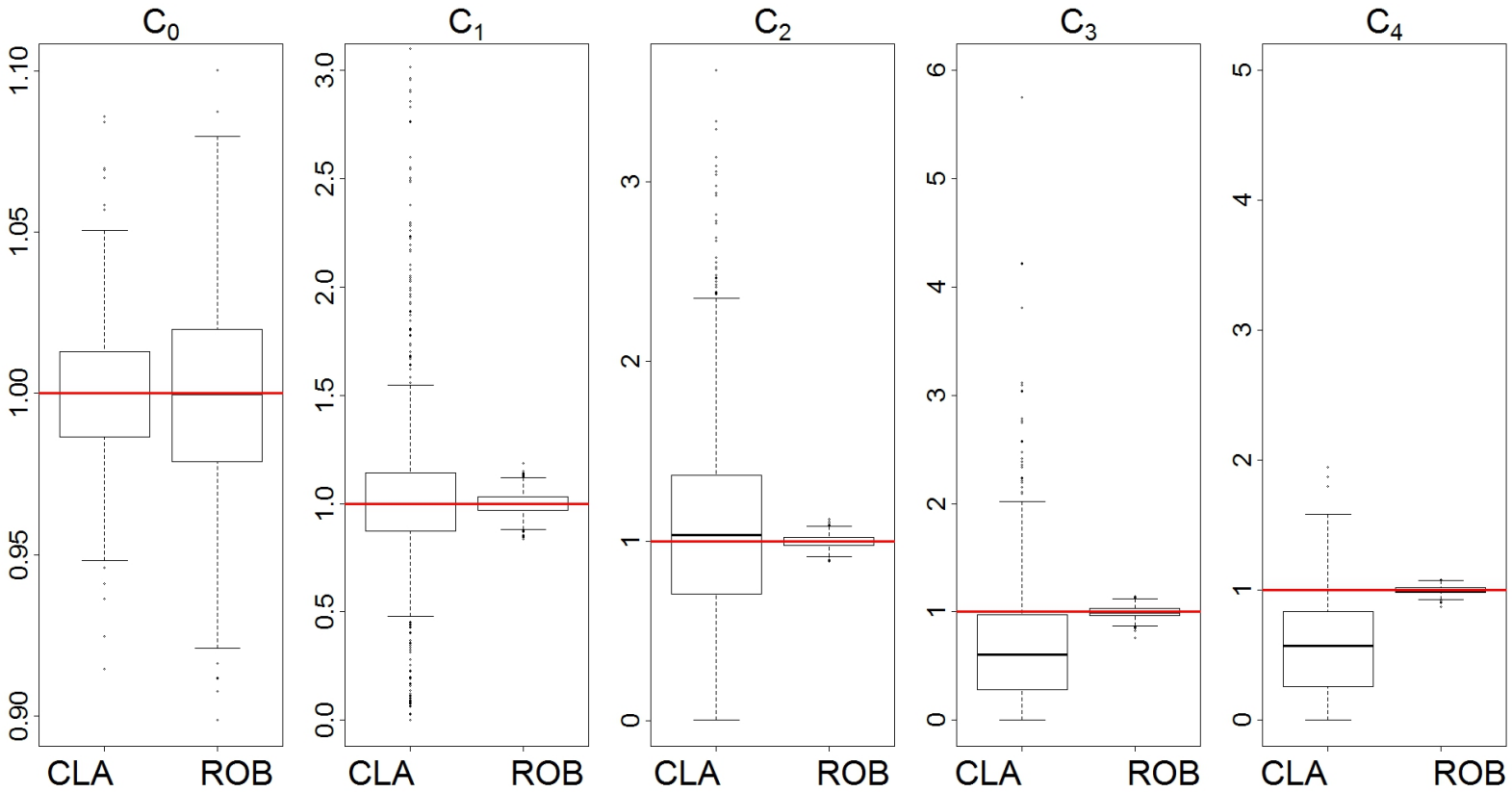
To measure the performance of the estimators of β_0,

- we compute the standard error (SD), the bias ($BIAS$) and the mean square error (MSE) for each component of β_0 over the $Nrep$ replications.

- we only show the boxplots and the density estimation of the estimators of β_0

Simulation Studies

Estimation of the first coordinate β_0



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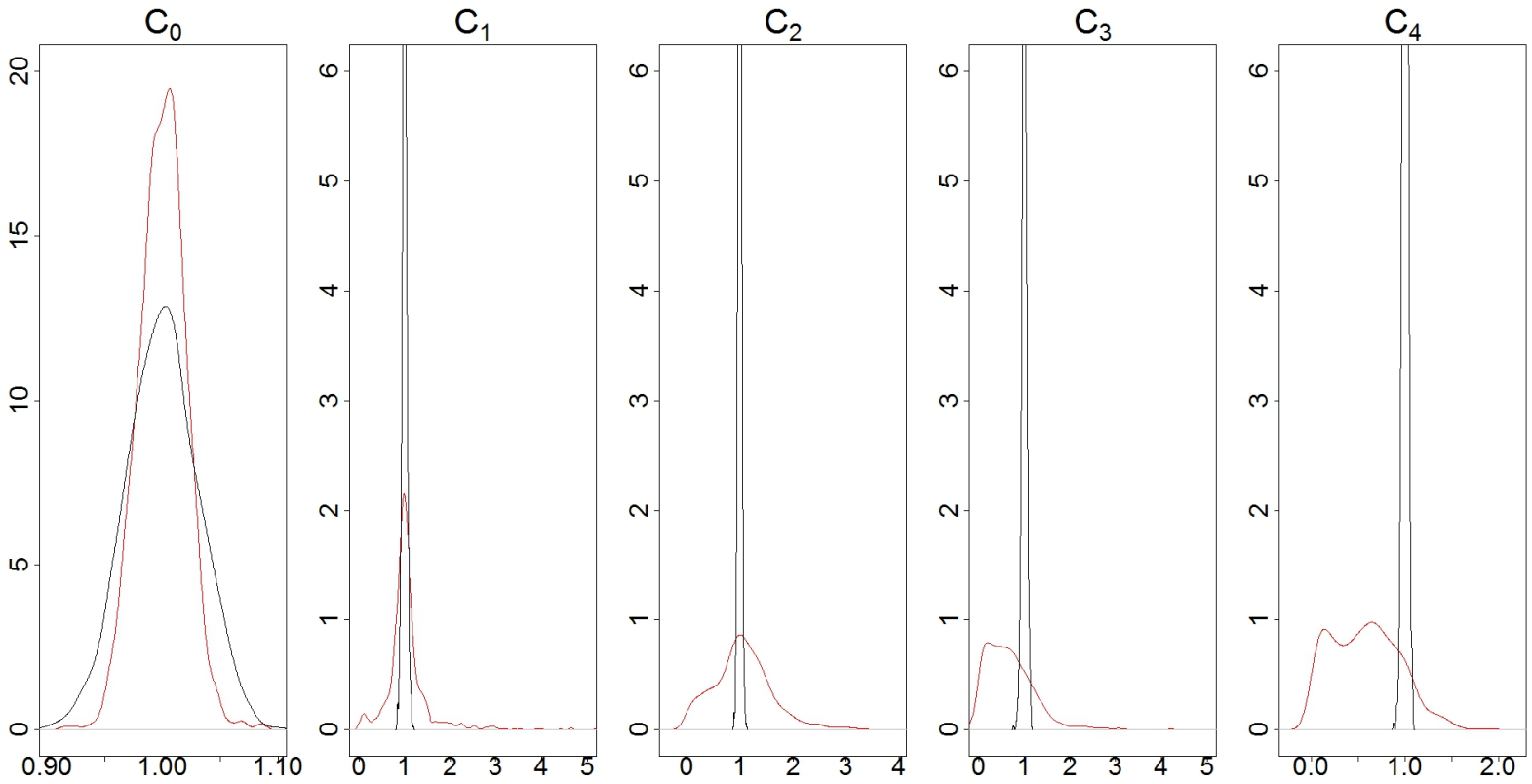
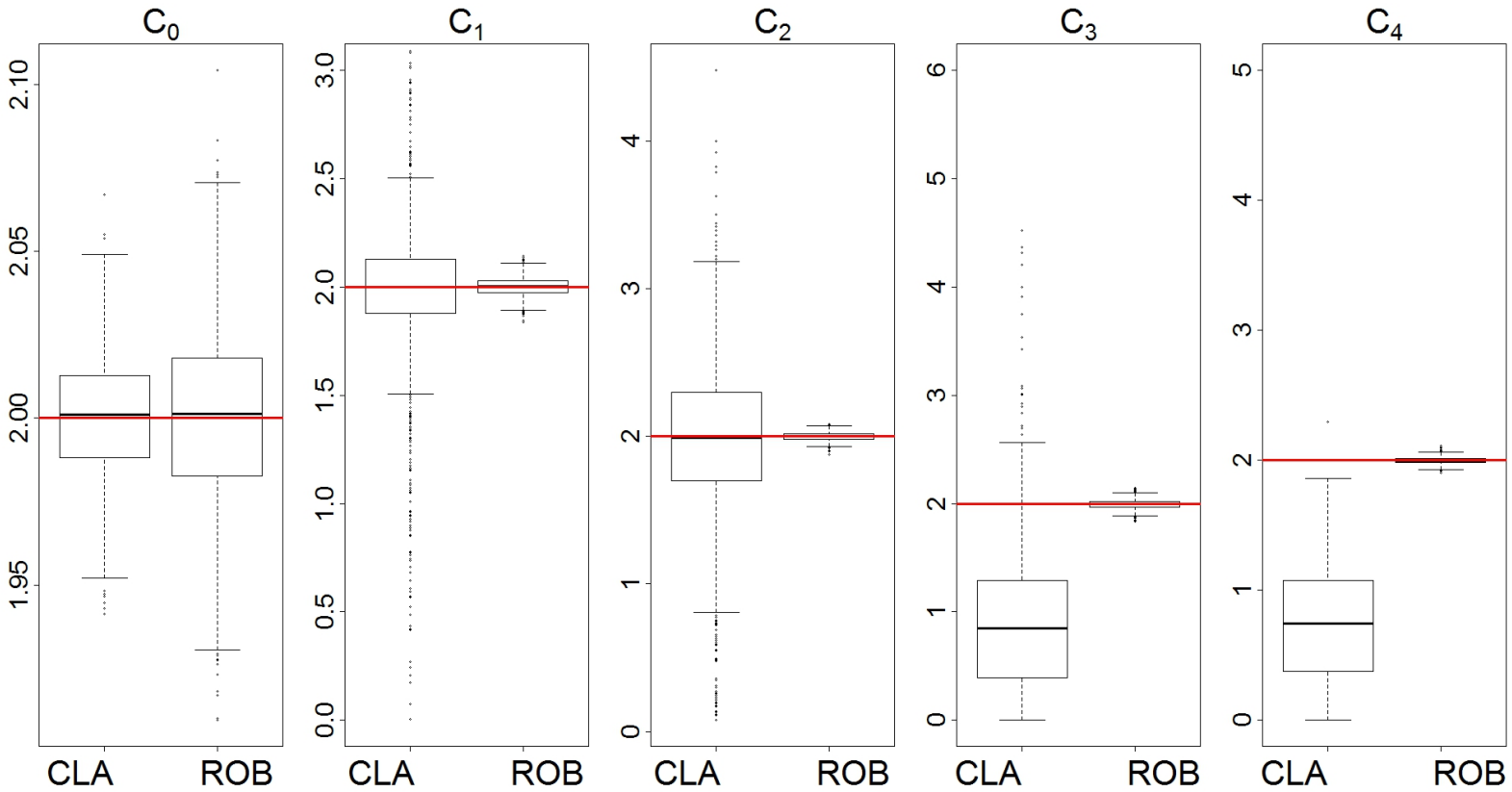


Simulation Studies

Estimation of the second coordinate β_0



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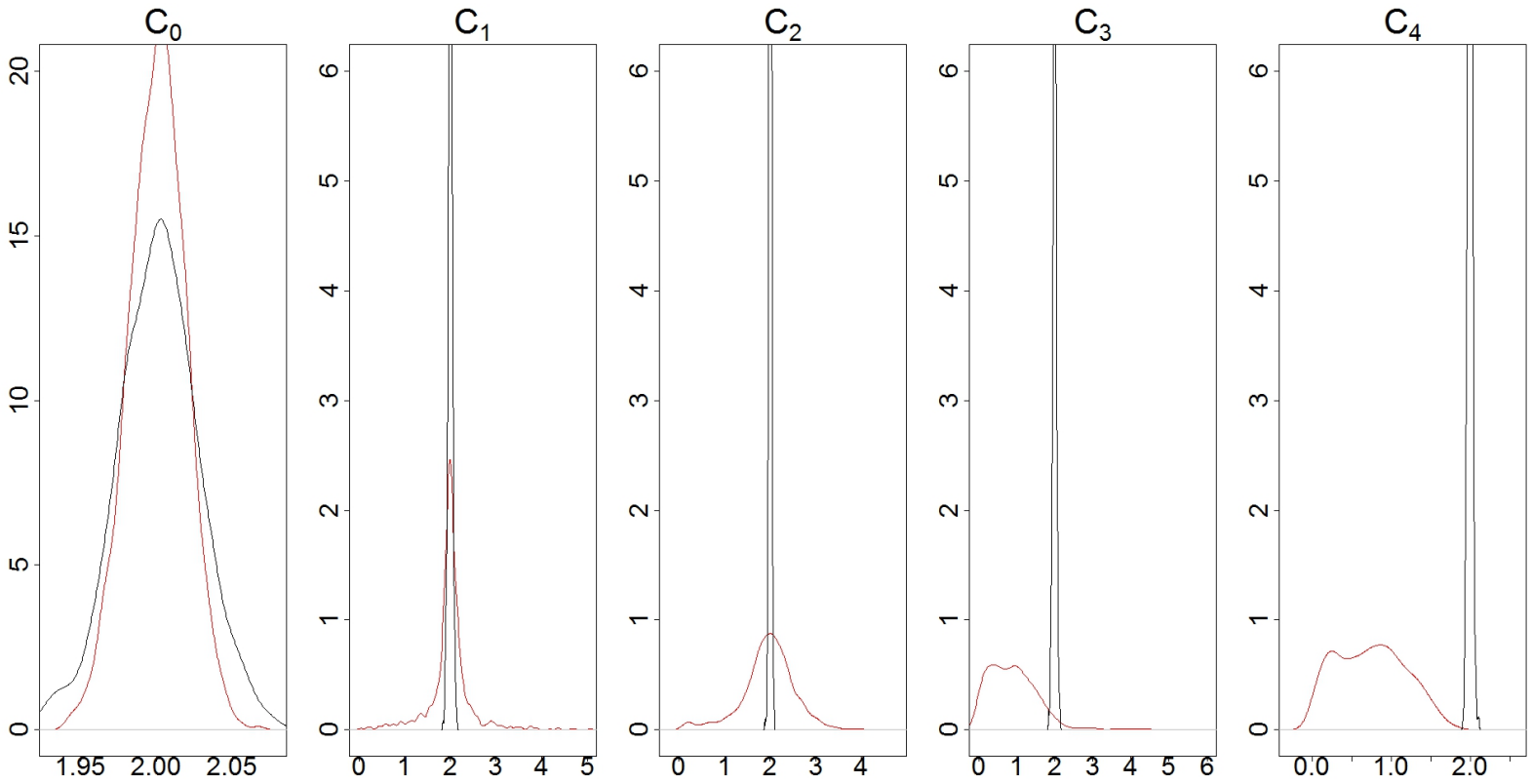
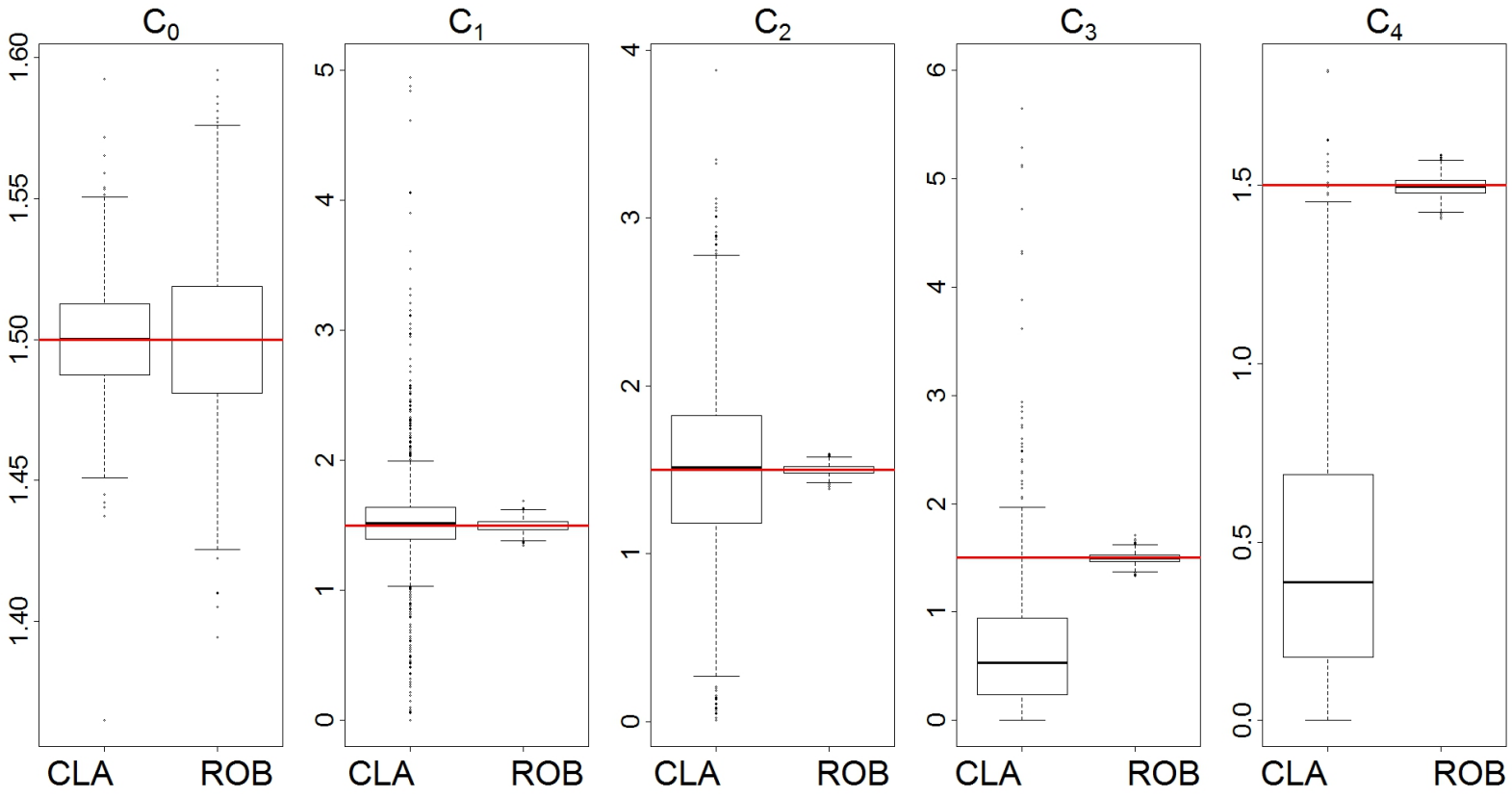


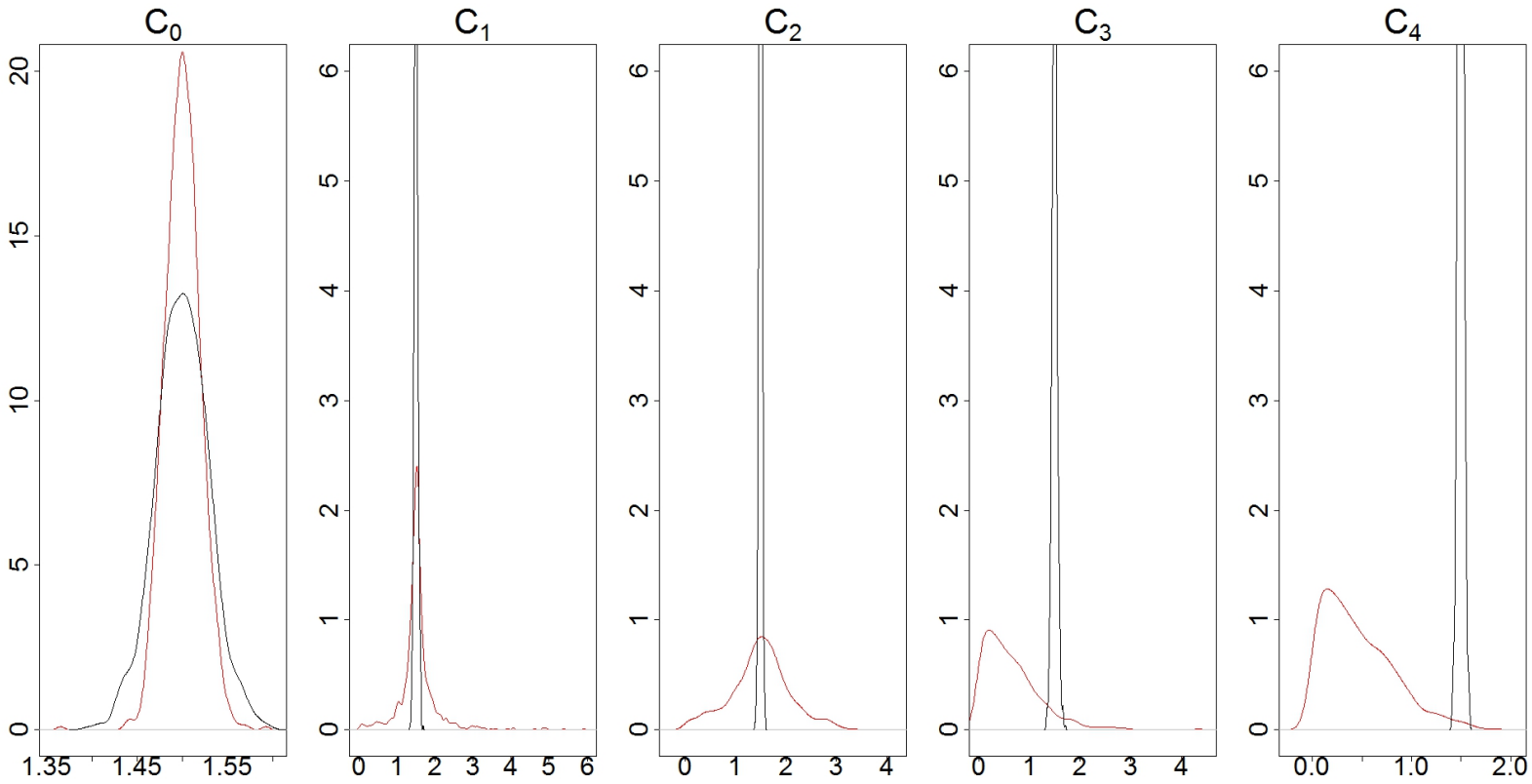
Simulation Studies

Estimation of the third coordinate β_0



Simulation Studies

Estimation of the third coordinate β_0



Simulation Studies

To measure the performance of the estimator of η_0, we compute the mean square error defined as follows

$$MSE(\widehat{\eta}) \quad = \quad \frac{1}{n}\sum_{i=1}^{n}(\widehat{\eta}(t_i) - \eta_0(t_i))^2.$$

and we consider its

mean ($M.MSE$)

median ($Med.MSE$)

standard error ($SD.MSE$)

Simulation Studies

		$M.MSE$	$SD.MSE$	$Med.MSE$
C_0	CLA	0.0926	0.0331	0.0869
	ROB	0.0997	0.0437	0.0923
C_1	CLA	5 62.9	8328	3.668
	ROB	0.1783	0.1275	0.1438
C_2	CLA	30.9236	35.0628	20.3381
	ROB	0.1084	0.0511	0.0981
C_3	CLA	573.5674	7893.1058	31.9639
	ROB	0.3012	0.2765	0.2257
C_4	CLA	31.5823	13.9275	29.4050
	ROB	0.1286	0.0651	0.1151

Air quality data set

The nonlinear model studied in Bianco and Spano is

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Note that, we omit the intercept to preserve the identifiability of the nonparametric function.
We compute the estimators using only the complete observations

Air quality data set

We consider the following estimators:

- a full robust nonparametric estimator (NP) using wind speed and temperature as explained variables (Boente and Fraiman(1989)).
- a robust Nonlinear estimator (NLR) using wind speed as covariable (Bianco and Spano (2017)).
- a robust partly Nonlinear model ($PNLR$) using wind speed and temperature as explained variables.

Air quality data set

We compute

$$
\begin{aligned}
ECM(\hat{m}) &= \frac{1}{n}\sum_{i=1}^{n}(O_i - \hat{m}(V_i, T_i))^2, && \text{the mean square error} \\
ECMed(\hat{m}) &= median((O_i - \hat{m}(V_i, T_i))^2), && \text{the median square error}
\end{aligned}
$$

where $\hat{m}(V, T)$ corresponds to the estimator of the regression function at (V, T).

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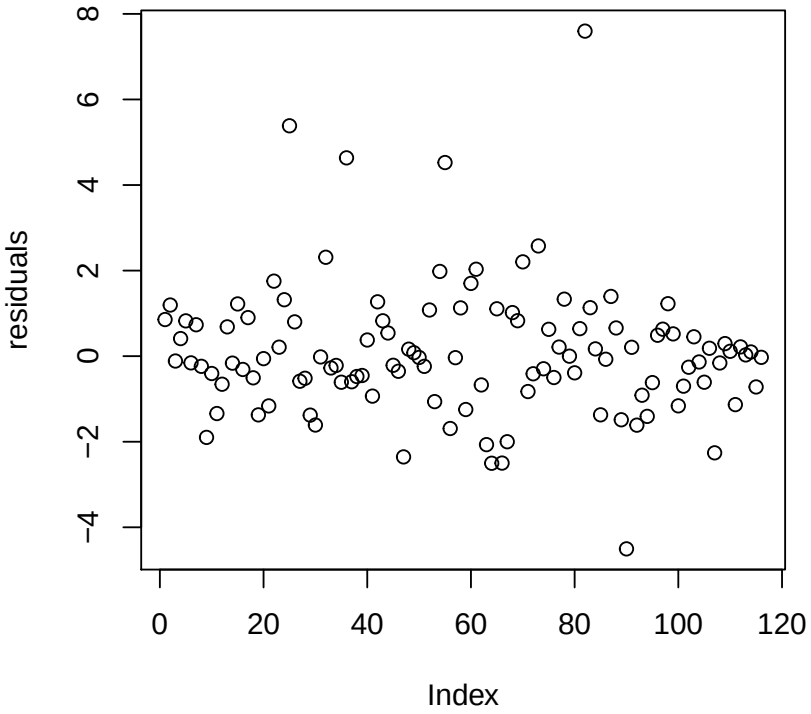
	$\hat{m}_{NP}$	$\hat{m}_{NLR}$	$\hat{m}_{PNLR}$
ECM	376.6914	903.6621	275.6815
$ECMed$	70.6501	126.3261	53.41841

Air quality data set

We use the robust estimates in order detect atypical observations.



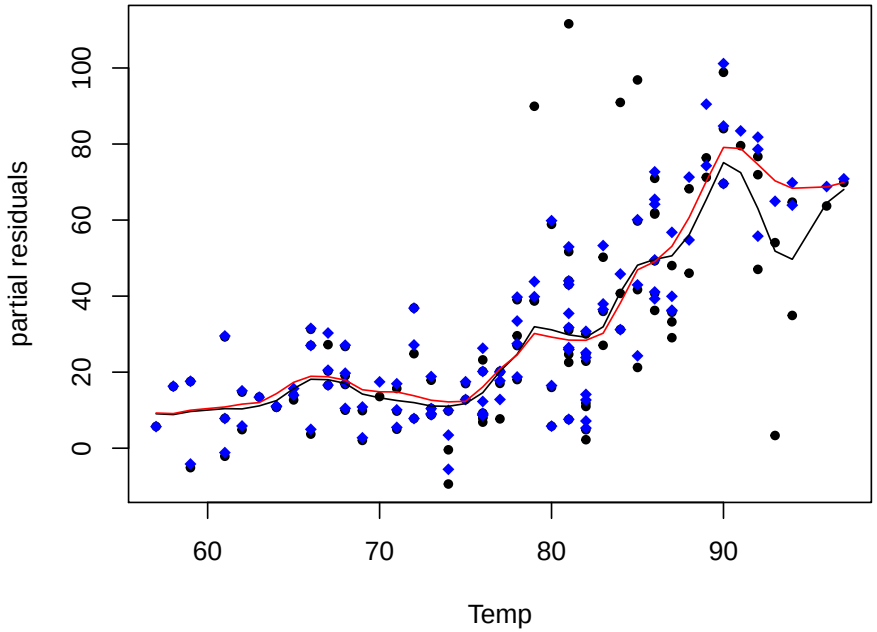
We can observed 5 outliers corresponding to observations 30, 62, 86, 117 and 126 in the sample.

Air quality data set

We delete these points of the sample and we compute the classic estimator. The robust method computed with whole sample performs similar than the classic estimators obtained with the clean sample.

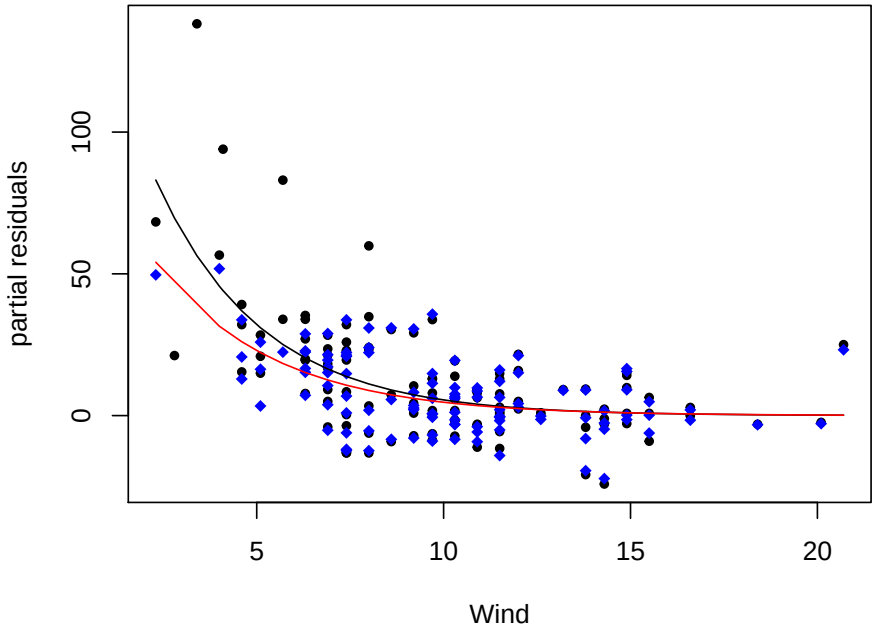
	$\hat{m}_{MC-5NP}$	$\hat{m}_{PNLR}$
ECM	125.4424	275.6815
ECMed	49.204	53.41841

Air quality data set



temperature vs

$res_i = O_i - g(\widehat{\beta}, V_i)$



wind speed vs

$\widetilde{res}_i = O_i - \widehat{\eta}(T_i)$

Concluding Remarks

- we proposed a robust estimation procedure for a partially nonlinear model.
- we derived asymptotic properties such as the consistency and the asymptotic normality.
- the simulation results support our findings about the asymptotic and robust properties of the estimation proposal.

Thank you for listening!

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On going or future research

- Make inference
- an automatic bandwidth selection
- estimation procedure taking into account missing responses